1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L33 – Experiment Design – I**

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4 **Empirical Study Stages**

- Broadly, four stages
  - Identification of research question(s)
  - Determination of variables
  - Design of experiment
  - Analysis of empirical data

5 **Understanding the Stages**

- We discussed the first stage (research question)
- In this lecture, we shall continue our discussion on remaining stages

6

Variables

7 **Basic Idea**

- Once research questions (or hypotheses) are framed, we identify the *variables*
  - To observe and record quantitative data

8 **Basic Idea**

- Consider RQ3 we discussed in the last lecture

RQ3: *Does the new interface let me enter text "faster" than MS Word*?

9 **Basic Idea**

- We want to observe text entry speed for RQ3
- So, text entry speed is a variable that takes values (real numbers) in the form of CPM

10 **Basic Idea**

- Text entry speed varies w.r.t. interface

- Another variable
- Takes interfaces as values (e.g., our design, MS Word, …)

11 **Basic Idea**

- Once we know variables and their values, we can set up test conditions by varying values systematically

12 **Basic Idea**

- Empirical study done based on these test conditions
- Therefore, identification of variables is necessary
- Along with variables, it is also important to *suitably define* their values

13

Measurement Scales

14 **Scales**

- In empirical research, we observe and record
- There are broadly TWO ways in which we can record the observations

15 **Scales**

- Manual recording - human observer records data
- We can also record data automatically without human observer – with technology such as computers, sensors, camera, logging software

16 **Scales**

- Recording of observed data involve measurement of the data

17 **Scales**

- We can measure using scales of measurement
- FOUR scales

18 **Scales**

- Nominal
  - We assign some (arbitrary) codes to attributes of observational data
  - E.g., suppose we wish to record gender data in an empirical study. Instead of recording data in terms of male or female, we may assign the code 1 to male and 2 to female and record the data in terms these two numbers
  - The term categorical data is also used

19 **Scales**

- Ordinal
  - We can assign some order on the observed data with this measurement scale
  - Ex- we may observe performance (in terms of playing a game) of three mobile phones and record our observation by ranking phones performance-wise as 1st, 2nd and 3rd

20 **Scales**

- Interval
  - Recording observations in terms of equally spaced values
  - E.g., If you see a thermometer closely, you might notice some closely spaced marker (lines) indicating specific values (e.g., 98°, 99° and so on). We record temperature based on these markings.

21 **Scales**

- Ratio
  - Uses ratio of two *quantities*
  - E.g., CPM - ratio of number of characters typed and minutes spent on typing
  - The most sophisticated scale of measurement

22 **Scales**

- Ordinal scales are heavily used for *rating*
- A popular rating scale is the Likert scale (named after its inventor who incidentally was a psychologist) [Likert, 1932]

23 **Scales**

- We should strive to use interval or ratio scale as much as possible
  - These scales support a wide array of analysis techniques as we shall see later

24 **Scales**

- Remember that the measurement scales are important for the observed variables only

25 **Scales**

- In RQ3, one variable is interface
- Which scale of measurement we used?
  - We may assign (arbitrarily) an integer (one) to each interface
  - That'd be using nominal scale

26 **Scales**

- We formed a research question having a variable that take nominal data
- Is that all right?
- Do we need to change the variable?

27 **Scales**

- Answer is NO – in fact, the questions are NOT RELEVANT

28 **Scales**

- The variable is used to generate test conditions - not to observe values
- We are observing text entry speeds
- Concern about measurement scales applies to speed variable only

29

Dependent & Independent Variables

30 **Variable Types**
- TWO types
  - Dependent variables
  - Independent variables

31 **Independent Variables**
- Let us reconsider RQ3
- We have considered two values for the "interface" variable
  - Our design
  - MS Word

32 **Independent Variables**
- We wish to observe text entry speed for each
  - Our design and for *MS Word*

33 **Independent Variables**
- How we arrived at these interfaces - we *assigned* some values to the variable
  - We *controlled* its values

34 **Independent Variables**
- Such variables, which are controlled, are known as *independent* variables
  - Sometimes (in fact, many a times) we use the term *factors* to denote these variables

35 **Independent Variables**
- The values assigned to variables are also known as *levels* (of the variable)
- In our example
  - Factor: interface
  - Levels (2): Our design, MS Word

36 **Independent Variables**
- Factors combined with their levels give us test conditions for the experiment
  - Ex - "*observe text entry speed for a given text with our design*"
- We can have TWO such test conditions for the factors and levels in our example

37 **Independent Variables**
- How do we set the levels for a factor - many ways
  - May come from prior knowledge and experience
  - Theoretical analysis of the problem
  - If all else fails, we might take recourse to a *pilot study* (a small-scale, typically informal and unstructured, study to prepare for the bigger study)

38 **Dependent Variables**
- In RQ3, there is also another variable – text entry rate (TER)
- TER depends on the interface (more specifically, the levels of the interface)
- Therefore, TER is called the dependent variable

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Control & ConfoundingVariables

40 **Control Variables**

- In controlled experiment, we control independent variables assuming those to be the only factors that influence observations
  - We may be wrong
- There may be other factors that we are not controlling, which can potentially affect observations

41 **Control Variables**

- Ex - we assumed only the interface factor affects text entry speed
- What about user gender - speed of a male user might differ from a female user, for the same interface
- A potential factor but we are not controlling it

42 **Control Variables**

- If we know gender is a factor, we may add a fourth factor, gender, to our list of factors

43 **Control Variables**

- There is another way of looking at it
  - We may not be interested to know the effect of gender on speed
  - In that case, we need not consider it as a factor

44 **Control Variables**

- Rather, we will use it as a control variable with a fixed value
  - For our experiment, we may use only the male participants
  - Therefore, we set the factor gender to male and use this setting for the whole experiment

45 **Control Variables**

- The only problem: any conclusion we draw will be applicable for the male participants only!

46 **Confounding Variables**

- Sometimes, we may not even be aware of the existence of factor(s) other than those we already identified (independent and control variables)
- A very common example is the factor called the *practice effect* (or the *learning effect*)

47 **Confounding Variables**

- In the experiment for RQ3, suppose a participant sees the two interfaces in a particular sequence
- After typing with first interface, participant gathers some experience and grows some expectations about nature of interfaces
- Experience and expectations potentially affects the way participant types with the second interface
- This is the practice effect

48 **Confounding Variables**

- Practice effect varies with test conditions systematically although we are not taking

into account this variation
- An example of confounding variables: we are either not aware of or do not take into consideration their existence in spite of them influencing observations
- We shall learn about a method to deal with the practice effect later

49 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
  - E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

Chapter 7